Explore Image Filtering Results

Copyright Barry Van Veen 2016

Contents

- Prepare Workspace
- Problem 1a
- Problem 1b
- Problem 1c
- Problem 2a
- Problem 2b
- Problem 2c
- Problem 3a
- Problem 3b

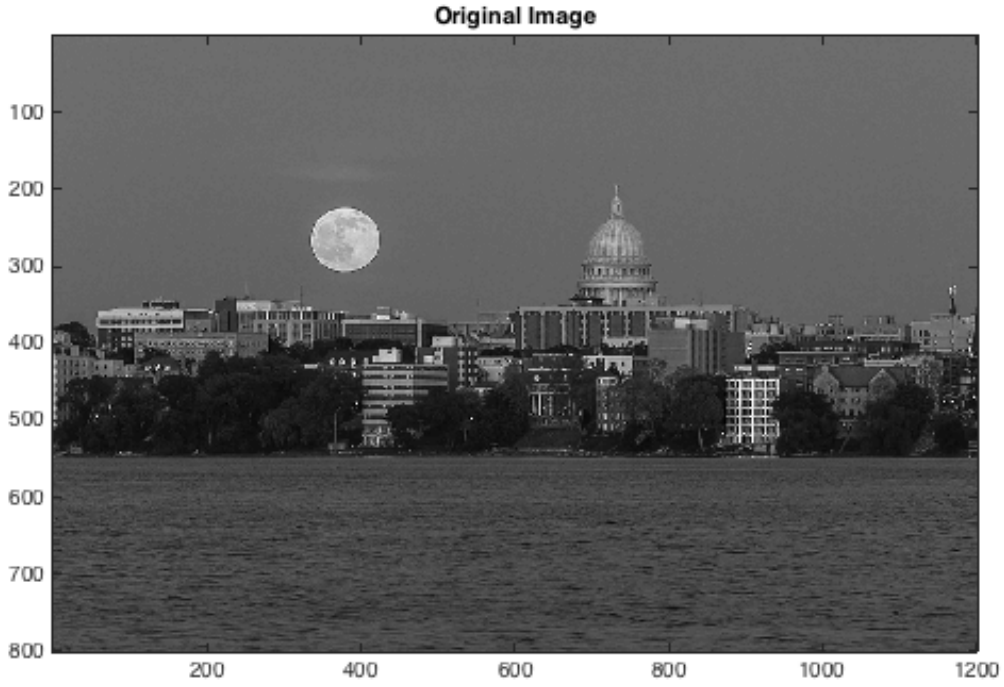
Prepare Workspace

```
clear
close all
clc
load Moon.mat % loads image named 'MadisonMoon'

im = double(MadisonMoon); % convert from uint8 to double precision format

% Display image

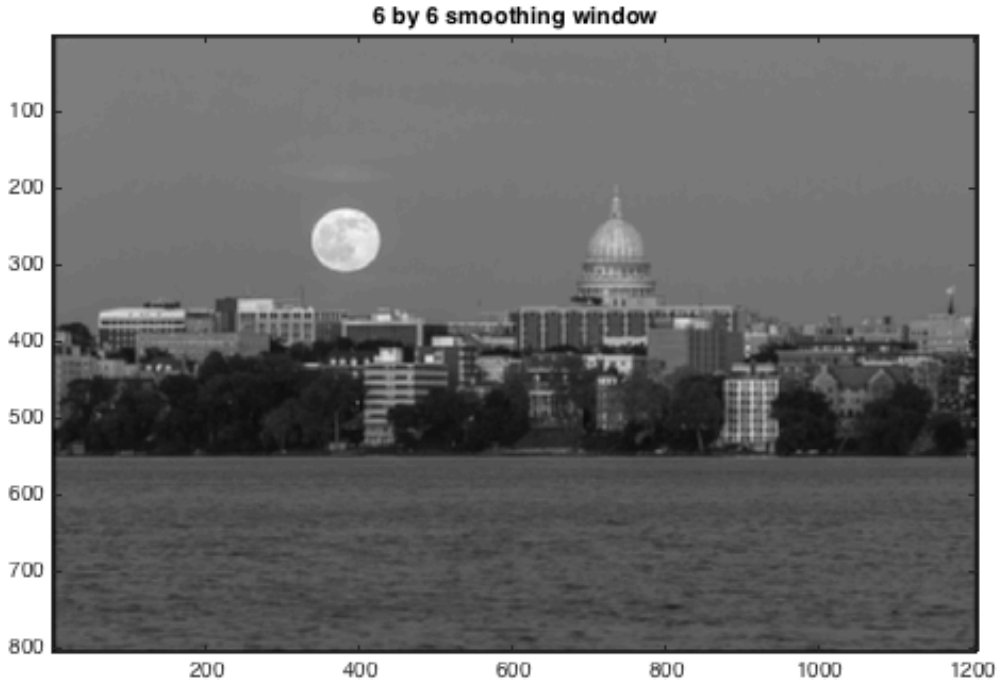
figure
imagesc(im)
axis equal
axis tight
colormap(gray)
title('Original Image')
```



Problem 1a

```
h_a = ones(6,6)/36;
ima = conv2(im,h_a);

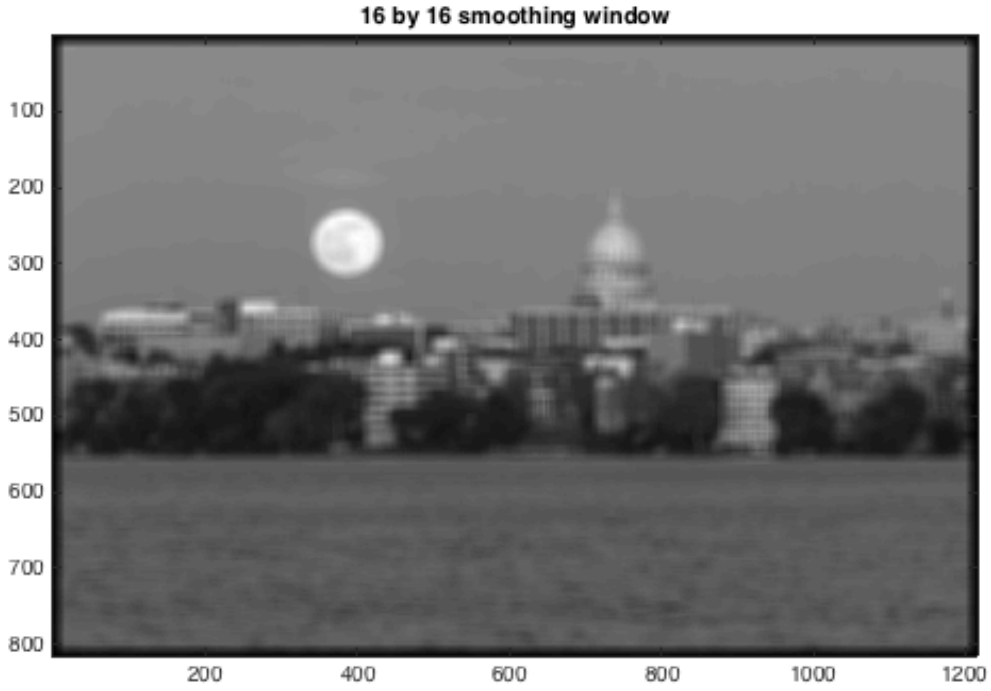
figure
imagesc(ima)
axis equal
axis tight
colormap(gray)
title('6 by 6 smoothing window')
```



Problem 1b

```
hb = ones(16,16)/256;
imb = conv2(im,hb);

figure
imagesc(imb)
axis equal
axis tight
colormap(gray)
title('16 by 16 smoothing window')
```



Problem 1c

```
hh = ones(1,6)/6;
hv = hh';

im_h = conv2(im,hh);
im_hv = conv2(im_h,hv);

figure
imagesc(im_hv)
axis equal
axis tight
colormap(gray)
title('Horiz then Vertically Blurred Image')

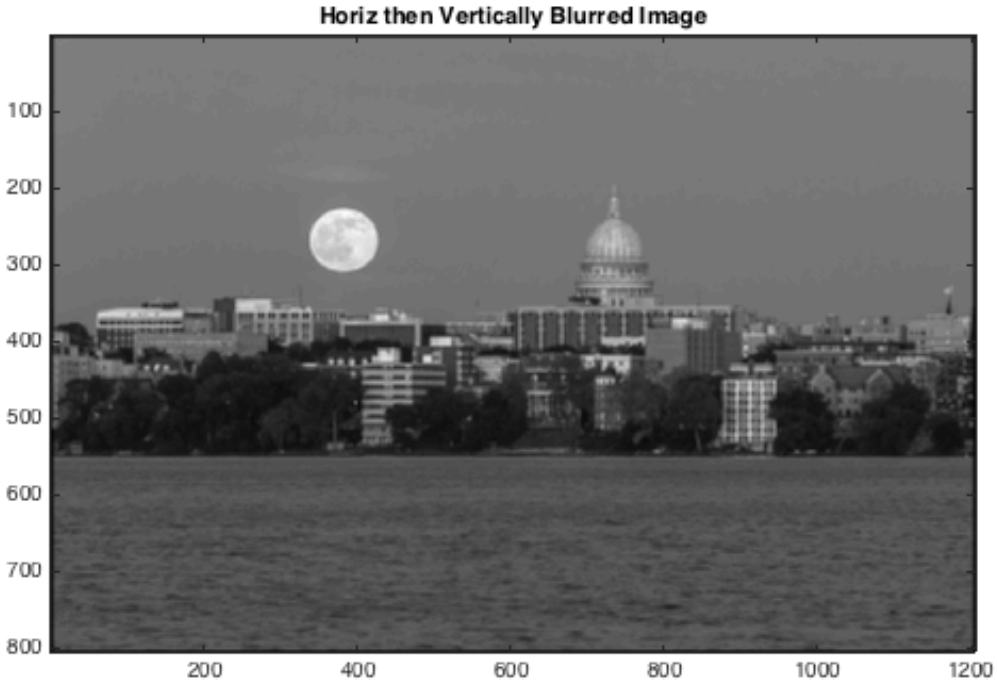
norm(ima-im_hv,'fro')

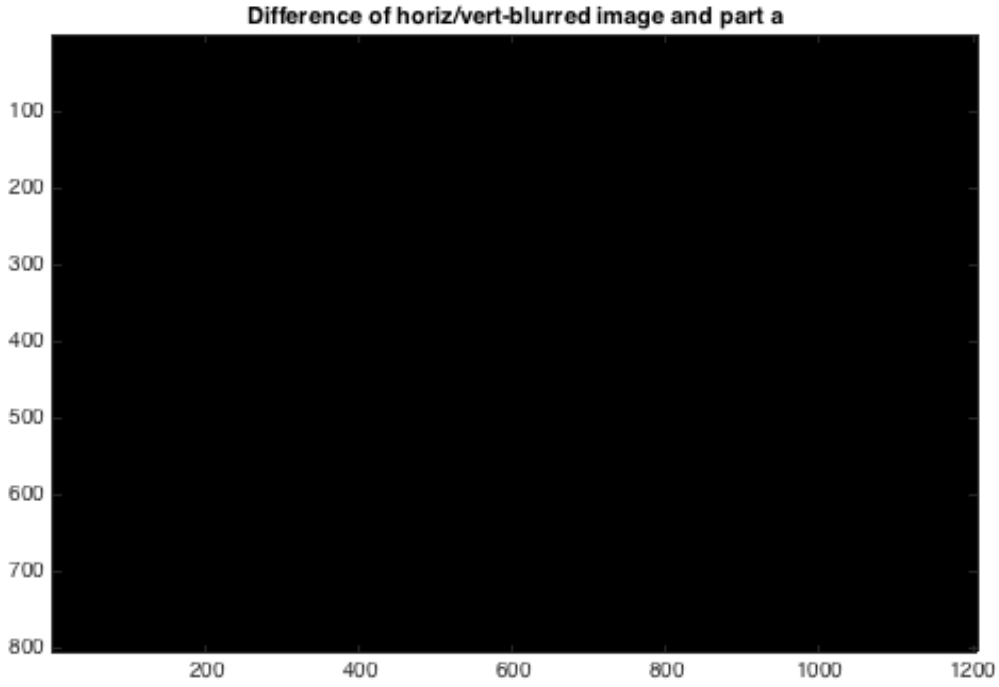
figure
image(ima-im_hv)
axis equal
axis tight
colormap(gray)
title('Difference of horiz/vert-blurred image and part a')

% Note that the image command was used instead of imagesc.  Imagesc is
% deceptive in this case since the differences between imb and imcc are on
% the order of 10^{-14}.
```

```
ans =

    1.7632e-11
```



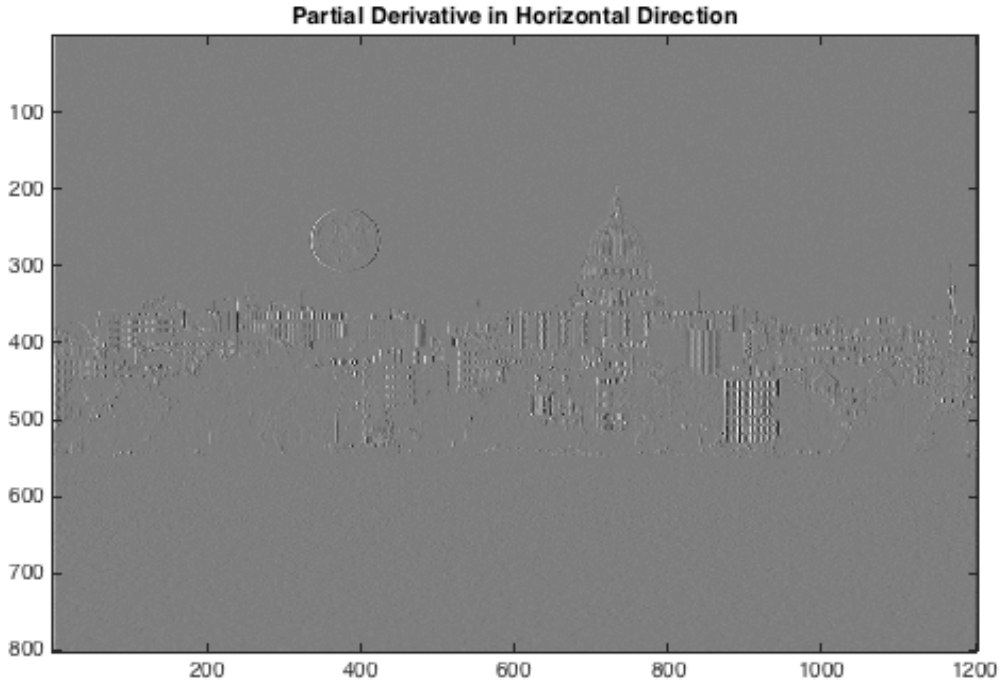


Problem 2a

```
hdh = [1 0 -1;2 0 -2;1 0 -1];

imhoriz = conv2(im,hdh);

figure
imagesc(imhoriz)
axis equal
axis tight
colormap(gray)
title('Partial Derivative in Horizontal Direction')
```

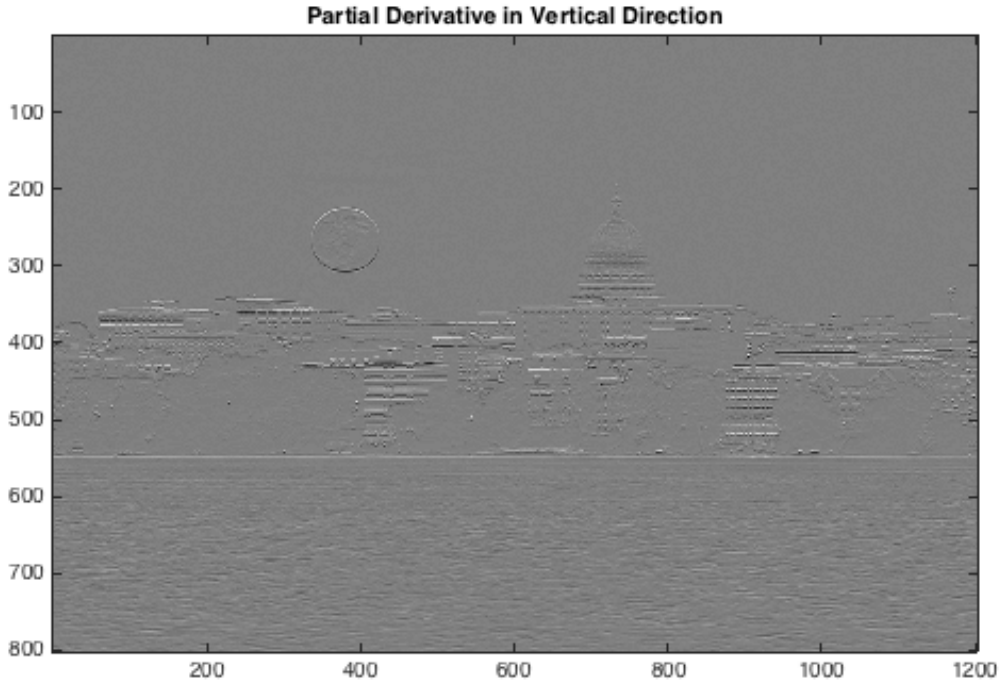


Problem 2b

```
hdv = [1 2 1;0 0 0;-1 -2 -1];

imvert = conv2(im,hdv);

figure
imagesc(imvert)
axis equal
axis tight
colormap(gray)
title('Partial Derivative in Vertical Direction')
```

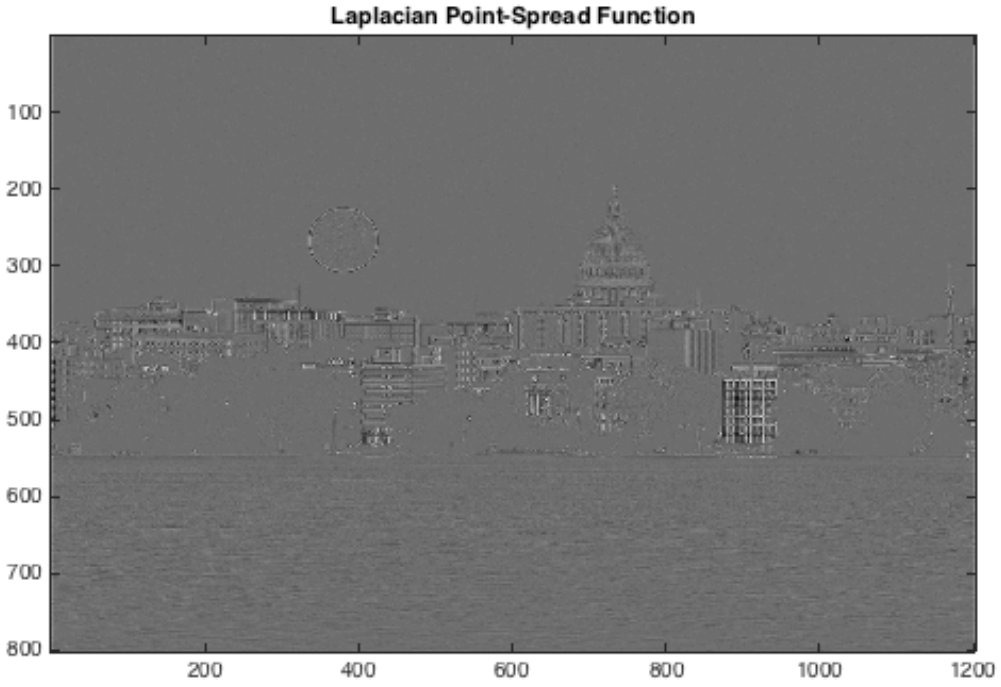


Problem 2c

Problem 3a

```
L = [-1 -1 -1; -1 8 -1; -1 -1 -1];

imL = conv2(im,L);
figure
imagesc(imL)
axis equal
axis tight
colormap(gray)
title('Laplacian Point-Spread Function')
```



Problem 3b

Published with MATLAB® R2015a